Chapter 1

Distributed Linear Algebra Infrastructure

In this document we provide an overview of oomph-lib's distributed linear algebra framework, discussing its design and key functionality. The aim of this framework is to facilitate the parallel (distributed) execution of linear algebra type operations with little (or no) user intervention. This requires all linear algebra data and computations to be distributed over all available processes.

We begin by defining the OomphCommunicator, a class that is fundamental to distributed computing in oomph-lib. Next we discuss the class LinearAlgebraDistribution which specifies the distribution of the data and computations over the available processes. In Sections DoubleVector, CRDoubleMatrix and DistributedLinearAlgebraObject we discuss oomph-lib's distributed linear algebra objects including the key containers (matrices and vectors) and operators (solvers and preconditioners). Finally, we demonstrate how the distributed linear algebra framework is typically used in practice.

The primary aim of this document is to provide an overview of the design and functionality of distributed linear algebra capabilities in oomph-lib, and hence we do not discuss every method of every class; we refer the reader to the class documentation for a complete specification.

1.1 OomphCommunicator

Oomph-lib employs MPI for distributed memory parallelisation. Fundamental to MPI is the communicator (MPI_Comm) which determines which processes are involved in a parallel computation. Although oomph-lib is implemented in C++, the C MPI bindings are utilised. Oomph-lib provides the class OomphCommunicator as an object-oriented wrapper for a MPI_Comm.

Calling MPI_Helpers::init(argc,argv) is equivalent to calling MPI_Init(argc,argv). It initialises MPI (i.e. calls MPI_Init(argc,argv)) and creates oomph-lib's global communicator:

```
//===start_of_main=====================================================
/// Driver code for generic mpi stuff
//=====================================================================
int main(int argc, char* argv[])
{

  // Initialise MPI
  MPI_Helpers::init(argc,argv);
```

The newly created communicator is available via a MPI_Helpers class static method:

```
  // Get the global oomph-lib communicator
  const OomphCommunicator* const comm_pt = MPI_Helpers::communicator_pt();
```

and is equivalent to the MPI_COMM_WORLD communicator in that it represents all the processes that oomph-lib knows about. By default, this communicator contains exactly the same set of processes as MPI_COMM_WORLD.

The OomphCommunicator provides a number of access functions to communicator data including the rank of the process and the number of the processes:

```
  // Get rank and total number of processors.
  unsigned my_rank = comm_pt->my_rank();
  unsigned nproc = comm_pt->nproc();

  // Tell us who you are...
  oomph_info « "I'm rank " « my_rank « " on a total of "
             « nproc « " processors" « std::endl;
```

1.2 LinearAlgebraDistribution

Distributed memory parallelisation requires data and computations to be distributed over the available processes in some way. In this document we are solely interested in distributed linear algebra. We choose to distribute the linear algebra objects row-wise, and constrain the distribution such that each process is associated with a single contiguous set of rows. The class `LinearAlgebraDistribution` allows the specification of such a distribution.

The distribution is defined by two integers defining the first global row and the number of local rows associated with a process. This data is sufficient to allow a mapping between a global row number and a local row number on a particular process.

To construct a `LinearAlgebraDistribution` in which 100 rows are uniformly distributed across the set of processes specified by `comm_pt` we write:

```
// Create a uniformly distributed LinearAlgebraDistribution
// 100 global rows are uniformly distributed accross the processes of
// comm_pt
unsigned nrow_global = 100;
LinearAlgebraDistribution distributed_distribution(comm_pt,
                                                   nrow_global);

// Show us how many rows this processor holds
oomph_info « "distributed distribution: first_row and nrow_local: "
           « distributed_distribution.first_row() « " "
           « distributed_distribution.nrow_local()
           « std::endl;
```

In this example, if run on four processes, the first 25 rows are associated with process 0, the next 25 rows are on process 1 and so on. In general, in a uniform distribution of n_r global rows over n_p processes the first row on process $p = 0, 1, \ldots, (n_p - 1)$ is $\lfloor \frac{pn_r}{n_p} \rfloor$. It is also possible the specify alternative user defined distributions; see the class documentation for details.

An optional third (`bool`) argument (default: true) in the constructor indicates that we require a distributed linear algebra distribution. However, on some occasions we may want to replicate all rows of a linear algebra object on all processes. This is achieved by simply making the third argument `false` (non-distributed):

```
// Construct an empty distribution object (does not specify a distribution)
LinearAlgebraDistribution locally_replicated_distribution;

// Build a locally replicated distribution such that every row is available
// on every process
bool distributed=false;
locally_replicated_distribution.build(comm_pt,nrow_global,distributed);

// Show us how many rows this processor holds
oomph_info « "locally replicated distribution: first_row and nrow_local: "
           « locally_replicated_distribution.first_row() « " "
           « locally_replicated_distribution.nrow_local()
           « std::endl;
```

This example illustrates two other features of `LinearAlgebraDistribution`. Firstly, the default constructor creates an empty distribution, and secondly for every (non-default) constructor there is an equivalent `build(...)` method to "re-construct" the object.

The state of the object is accessible through a range of methods.

```
// Get the number of local rows on this process
unsigned nrow_local = distributed_distribution.nrow_local();

// Get the first row on this process
unsigned first_row = distributed_distribution.first_row();

// Get the number of global rows
nrow_global = distributed_distribution.nrow();

// Is this distributed (true) or locally replicated (false)
distributed = distributed_distribution.distributed();

// Does this object specify a distribution
bool built = distributed_distribution.built();
```

The `built()` method indicates if the object specifies a distribution, or is empty.

1.3 DoubleVector

The simplest distributed linear algebra object is `DoubleVector`, a distributed vector of `doubles` developed specifically for linear algebra (It differs from a `Vector<double>` which simply provides a container for `doubles`). For example, the following command constructs a `DoubleVector` with a uniform distribution (specified by the distributed `LinearAlgebraDistribution` defined in the previous section) and unit elements:

```
// Construct a uniformly distributed DoubleVector with unit elements
DoubleVector my_vector(distributed_distribution,1.0);
```

To access the vector elements the `operator[]` is implemented. For example to increment every element by one:

```
// Increment every element of my_vector on this process by 1
nrow_local = my_vector.distribution_pt()->nrow_local();
for (unsigned i = 0; i < nrow_local; i++)
 {
  my_vector[i]+=1.0;
 }
```

It is the `oomph-lib` convention that the data in `DoubleVector` (and all other distributed linear algebra object) is accessed using local indices. The following loop documents the local row number, the global row number, and the value of the elements on each process:

```
// Document elements on this process in my_vector
nrow_local = my_vector.distribution_pt()->nrow_local();
first_row = my_vector.distribution_pt()->first_row();
for (unsigned i = 0; i < nrow_local; i++)
 {
  oomph_info << "local row " << i
             << " is global row " << first_row+i
             << " and has value " << my_vector[i] << std::endl;
 }
```

To change the distribution of a `DoubleVector` while retaining the data, we provide the `redistribute(...)` method. For example to change `my_vector` from uniformly distributed to locally replicated:

```
// Redistribute my_vector such that it is locally replicated on all processes
my_vector.redistribute(&locally_replicated_distribution);
```

Just like the `LinearAlgebraDistribution`, we provide `build()` methods that mirror the behaviour of all non-default constructors. For example to revert `my_vector` to a uniform distribution with unit elements:

```
// (Re)build my_vector such that it is uniformly distributed over all
// processes
my_vector.build(distributed_distribution,1.0);
```

It is important to differentiate between `build(...)` and `redistribute(...)`; calling `build(...)` deletes the existing data, effectively re-constructing the object, whereas `redistribute(...)` retains the vector's data.

Like the `LinearAlgebraDistribution`, a `DoubleVector` need not contain any data. To generate an object in this state, we could instantiate an object using the default constructor or call the `clear()` method:

```
// Construct an empty DoubleVector
DoubleVector another_vector;

// Clear all data from an existing DoubleVector
my_vector.clear();
```

Again the `built()` method returns the state of the object and indicates if it contains any data.

1.4 CRDoubleMatrix

`CRDoubleMatrix` is the only distributed matrix in `oomph-lib`. It employs sparse compressed row storage to store `double` coefficients.

A `CRDoubleMatrix` has three fundamental states:

- A `CRDoubleMatrix` can have no distribution or coefficients in which case `my_matrix->distribution_built()` and `my_matrix->built()` are both `false`.
- A (built) distribution but no coefficients in which case `my_matrix->distribution_built()` is `true` but `my_matrix->built()` is still `false`.
- A (built) distribution and coefficients in which case `my_matrix->distribution_built()` and `my_matrix->built()` are both `true`.

For example, to construct an empty matrix we call:

```
// Construct an empty CRDoubleMatrix
CRDoubleMatrix my_matrix;
```

To specify the distribution as defined by the `LinearAlgebraDistribution distributed_distribution` we write:

```
// Specify that the rows be uniformly distributed
my_matrix.build(&distributed_distribution);
```

The distribution has now been specified but the coefficients have not. Like the `DoubleVector`, rows are indexed locally and hence the coefficients rows must be indexed locally. For example, to populate `my_matrix` as a square identity matrix, we write:

```
// Vector of coefficient of value 1.0
Vector<double> values(nrow_local,1.0);

// Column indices corresponding to values
Vector<int> column_indices(nrow_local);

// Index of vectors values and column_indices where the i-th row starts
// (each row contains one coefficient)
Vector<int> row_start(nrow_local+1);

// populate column_indices and row_start
for (unsigned i = 0; i < nrow_local; ++i)
 {
  column_indices[i]=first_row+nrow_local;
  row_start[i]=i;
 }
row_start[nrow_local]=nrow_local;

// Build the (square) matrix
unsigned ncol = nrow_global;
my_matrix.build(ncol,values,column_indices,row_start);
```

We note that the column indices are global because only the rows are distributed. The assembly of a `CRDoubleMatrix` is now complete.

We constructed the matrix in two stages by first specifying the distribution and then specifying the coefficients. However it is possible to perform this operation in just one step, by using the appropriate constructor or `build(...)` method, for example:

```
CRDoubleMatrix my_matrix2(&distributed_distribution,ncol,values,
                          column_indices,row_start);
```

1.5 DistributedLinearAlgebraObject

In this section we introduce the class `DistributedLinearAlgebraObject`, a base class for all distributed linear algebra objects. This class encapsulates a `LinearAlgebraDistribution`, provides (protected) access to derived classes to update (`build_distribution(...)`) and clear (`clear()`) the stored distribution. Secondly, it provides methods to simplify access to commonly used `LinearAlgebraDistribution` data. For example, because a `CRDoubleMatrix` is a `DistributedLinearAlgebraObject`,

```
// Get the first (global) row of my_matrix on this process
first_row =  my_matrix2.distribution_pt()->first_row();
```

can be replaced with

```
// Get the first (global) row of my_matrix on this process
first_row =  my_matrix2.first_row();
```

`DistributedLinearAlgebraObjects` can be divided into two types: containers and operators. We have already reviewed the containers `DoubleVector` and `CRDoubleMatrix`. A wide range of operator classes have been implemented in `oomph-lib` to operate on these containers. In particular, all `Linear↩ Solvers`, `IterativeLinearSolvers` and `Preconditioners` (discussed in the Linear Solvers Tutorial) are `DistributedLinearAlgebraObjects`. We finish this section by reviewing the key linear algebra operators:

- `SuperLUSolver` is a `LinearSolver` wrapper to both the SuperLU direct solver and the SuperLU Dist distributed direct solver. By default, whenever possible this class will automatically perform distributed solves.
- `TrilinosAztecOOSolver` is an `IterativeLinearSolver` wrapper to the Trilinos AztecOO package implementation of distributed Krylov methods including CG, GMRES and BiCGStab.
- `TrilinosMLPreconditioner` is a wrapper to the distributed Trilinos ML AMG preconditioners.
- `TrilinosIFPACKPreconditioner` is a wrapper to the distributed Trilinos IFPACK preconditioners.
- `HyprePreconditioner` is a wrapper to the distributed Hypre Scalable Linear Solvers package, of particular interest is the classical AMG implementation `BoomerAMG`.
- `MatrixVectorProduct` is a wrapper to the Trilinos Epetra distributed matrix-vector product implementation.

1.6 Distributed Linear Algebra In Practice

Having discussed `oomph-lib's` linear algebra infrastructure, we finally remark that `oomph-lib` is implemented such that linear algebra in `oomph-lib` is automatically distributed if executed under MPI on multiple processes. Specifically, a user should not need to specify either a `LinearAlgebraDistribution` or a `OomphCommunicator`, unless they wish to customise some aspect of the parallelisation.

All functionality is designed such that if a user does not specify a `LinearAlgebraDistribution`, then as much data and computation as possible will be uniformly distributed over all available processes.

As an example, we consider the `Problem` method `get_jacobian(...)`. If the user does not specify a return distribution for the Jacobian and residuals, then `oomph-lib` will uniformly distribute both containers.

```
// Set up a problem:
// Solve a 1D Poisson problem using a source function that generates
// a fish shaped exact solution
```

```
unsigned n_element=40;
OneDPoissonProblem<QPoissonElement<1,4> >
 problem(n_element,FishSolnOneDPoisson::source_function);

// Get the residual and Jacobian, by default both are uniformly distributed
// over all available processes
my_vector.clear();
my_matrix.clear();

// Get the Jacobian
problem.get_jacobian(my_vector,my_matrix);

oomph_info
 << "Uniformly distributed residual vector: first_row and nrow_local: "
 << my_vector.first_row() << " "
 << my_vector.nrow_local() << std::endl
 << "Uniformly distributed jacobian matrix: first_row, nrow_local and nnz: "
 << my_matrix.first_row() << " "
 << my_matrix.nrow_local() << " "
 << my_matrix.nnz() << " "
 << std::endl;
```

On the other hand, a user can specify a return distribution by setting the distribution of the matrix and vector prior to calling `get_jacobian(...)`.

```
// Request locally replicated residual and Jacobian from the problem
distributed=false;
LinearAlgebraDistribution locally_replicated_distribution_for_jac(
 comm_pt,problem.ndof(),distributed);

// Show us how many rows this processor holds
oomph_info
 << "locally replicated distribution_for_jac: first_row and nrow_local: "
 << locally_replicated_distribution_for_jac.first_row() << " "
 << locally_replicated_distribution_for_jac.nrow_local()
 << std::endl;
```

We finally remark that because all linear algebra operations are automatically distributed, to parallelise `oomph-lib's` Newton solve phase, the user need only run their executable under MPI on multiple processes.

1.7 Source files for this tutorial

- The source files for this tutorial are located in the directory:

  demo_drivers/self_test/mpi/generic_mpi

- The driver code is:

  demo_drivers/self_test/mpi/generic_mpi/generic_mpi_test.cc

1.8 PDF file

A pdf version of this document is available. \